Introduction to Radiobiological Models

A short story

Souvik Das

Resisdent Medical Physicist
Tata Memorial Hospital, Mumbai

Table of Contents

Table of Contents

Target theory

- The target of radiation for the purpose of therapy is the **DNA**.

Target theory

- The target of radiation for the purpose of therapy is the **DNA**.
- Post-irradiation **cell survival** as a function of radiation dose can be **biophysically modeled** in several different ways.

Target theory

- The target of radiation for the purpose of therapy is the **DNA**.
- Post-irradiation **cell survival** as a function of radiation dose can be **biophysically modeled** in several different ways.
- **Poisson statistics** are used for the basis for survival equations.

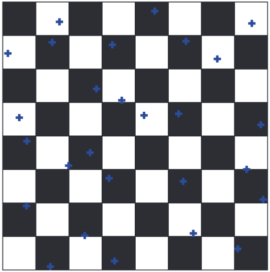


Figure: If it rains on a chessboard, and an average of x drops hits each square, How many squares are dry?

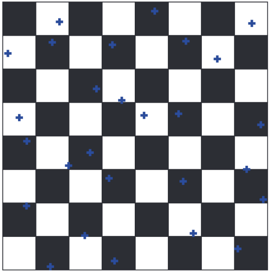


Figure: If it rains on a chessboard, and an average of x drops hits each square, How many squares are dry?

- Poisson statistics describe a large number of random events happening to a large number of subjects, averaging out to a small number of events per subject.

- If x is the average number of drops per square, then the probability of a square being hit by k drops is given by:

$$P(k) = \frac{x^k e^{-x}}{k!}$$

- If x is the average number of drops per square, then the probability of a square being hit by k drops is given by:

$$P(k) = \frac{x^k e^{-x}}{k!}$$

- The probability of a square being dry is given by:

$$P(0) = e^{-x}$$

Mathematical Modeling

Poisson Statistics

- If x is the average number of drops per square, then the probability of a square being hit by k drops is given by:

$$P(k) = \frac{x^k e^{-x}}{k!}$$

- The probability of a square being dry is given by:

$$P(0) = e^{-x}$$

- The probability of a square being wet is given by:

$$P(\text{wet}) = 1 - e^{-x}$$

- The Poisson model is used for surviving fraction (**SF**) of cells:

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:
    - e^{-x} cells survive (no hits).

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,

Poisson Distribution
Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 hit kills 63% of cells. SF $= 0.37$

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 hit kills 63% of cells. $\text{SF} = 0.37$
    - 2 hits kills 86% of cells. $\text{SF} = 0.14$

Poisson Distribution
Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 hit kills 63% of cells. SF $= 0.37$
    - 2 hits kills 86% of cells. SF $= 0.14$
    - 3 hits kills 95% of cells. SF $= 0.05$

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 hit kills 63% of cells. SF $= 0.37$
    - 2 hits kills 86% of cells. SF $= 0.14$
    - 3 hits kills 95% of cells. SF $= 0.05$
    - 4 hits kills 98% of cells. SF $= 0.02$

Poisson Distribution
Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 hit kills 63% of cells. SF $= 0.37$
    - 2 hits kills 86% of cells. SF $= 0.14$
    - 3 hits kills 95% of cells. SF $= 0.05$
    - 4 hits kills 98% of cells. SF $= 0.02$
    - 5 hits kills 99% of cells. SF $= 0.01$

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
  - At an average of x lethal hits per cell:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 hit kills 63% of cells. SF = 0.37
    - 2 hits kills 86% of cells. SF = 0.14
    - 3 hits kills 95% of cells. SF = 0.05
    - 4 hits kills 98% of cells. SF = 0.02
    - 5 hits kills 99% of cells. SF = 0.01

Note

D_0 is defined as the radiation dose resulting in 37% survival. This is assumed to be equal to the radiation dose necessary to cause one **lethal** hit per cell.

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
    - At an average of x surviving tumor cells per patient:

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
  - At an average of x surviving tumor cells per patient:
    - e^{-x} cells survive (no hits).

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
  - At an average of x surviving tumor cells per patient:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).

Poisson Distribution
Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
  - At an average of x surviving tumor cells per patient:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
  - At an average of x surviving tumor cells per patient:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 tumor cell per pt.: TCP $= 0.37$

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
  - At an average of x surviving tumor cells per patient:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 tumor cell per pt.: TCP $= 0.37$
    - 2 tumor cell per pt.: TCP $= 0.14$

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
  - At an average of x surviving tumor cells per patient:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 tumor cell per pt.: $\text{TCP} = 0.37$
    - 2 tumor cell per pt.: $\text{TCP} = 0.14$
    - 3 tumor cells per pt.: $\text{TCP} = 0.05$

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
  - At an average of x surviving tumor cells per patient:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 tumor cell per pt.: $\text{TCP} = 0.37$
    - 2 tumor cell per pt.: $\text{TCP} = 0.14$
    - 3 tumor cells per pt.: $\text{TCP} = 0.05$
    - 4 tumor cells per pt.: $\text{TCP} = 0.02$

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
  - At an average of x surviving tumor cells per patient:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 tumor cell per pt.: TCP $= 0.37$
    - 2 tumor cell per pt.: TCP $= 0.14$
    - 3 tumor cells per pt.: TCP $= 0.05$
    - 4 tumor cells per pt.: TCP $= 0.02$
    - 5 tumor cells per pt.: TCP $= 0.01$

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
  - At an average of x surviving tumor cells per patient:
    - e^{-x} cells survive (no hits).
    - $1 - e^{-x}$ cells die (at least 1 hit).
  - Based on this equation,
    - 1 tumor cell per pt.: TCP $= 0.37$
    - 2 tumor cell per pt.: TCP $= 0.14$
    - 3 tumor cells per pt.: TCP $= 0.05$
    - 4 tumor cells per pt.: TCP $= 0.02$
    - 5 tumor cells per pt.: TCP $= 0.01$

Note

To achieve a certain TCP, you should aim for a tumor cell survival of (1 - TCP).

Target theory
Single-Target, Single-Hit Model

- It assumes that each cell has a **single critical target**, and that a single hit to this target is sufficient to cause cell death.

Target theory

Single-Target, Single-Hit Model

- It assumes that each cell has a **single critical target**, and that a single hit to this target is sufficient to cause cell death.
- The survival fraction (SF) is given by:

$$SF = e^{-D/D_0}$$

where D is the dose and D_0 is the dose required to reduce the surviving fraction to 37%.

Target theory
Single-Target, Single-Hit Model

- It assumes that each cell has a **single critical target**, and that a single hit to this target is sufficient to cause cell death.
- The survival fraction (SF) is given by:

$$SF = e^{-D/D_0}$$

  where D is the dose and D_0 is the dose required to reduce the surviving fraction to 37%.
- This type of survival curve is seen for mammalian cells which are irradiated with **high LET** radiation such as alpha particles or carbon ions; very **radiation-sensitive** cells like lymphocytes and bone marrow cells; cells that have major defects in DNA doublestrand break repair; are synchronized in M phase; and for cells that have DNA double-strand break repair inhibited by chemical or gene knockdown.

Target theory

Multitarget, Single-Hit Model

- This model assumes that each cell has multiple independent targets, all of which must be hit to kill the cell.

Target theory

Multitarget, Single-Hit Model

- This model assumes that each cell has multiple independent targets, all of which must be hit to kill the cell.
- For a single dose D, the survival fraction (SF) is given by:

$$SF = 1 - (1 - e^{-D/D_0})^n$$

where D is the dose, D_0 is the dose required to reduce the surviving fraction to 37%, and n is the number of targets.

Target theory

Multitarget, Single-Hit Model

- This model assumes that each cell has multiple independent targets, all of which must be hit to kill the cell.
- For a single dose D, the survival fraction (SF) is given by:

$$SF = 1 - (1 - e^{-D/D_0})^n$$

  where D is the dose, D_0 is the dose required to reduce the surviving fraction to 37%, and n is the number of targets.
- This type of survival curve is seen for mammalian cells which are irradiated with **low LET** radiation such as X-rays or gamma rays; cells that are **radiation-resistant** like fibroblasts and epithelial cells; cells that have a normal DNA double-strand break repair; and for cells that are synchronized in G1 or S phase.

Target theory

Multitarget, Single-Hit Model

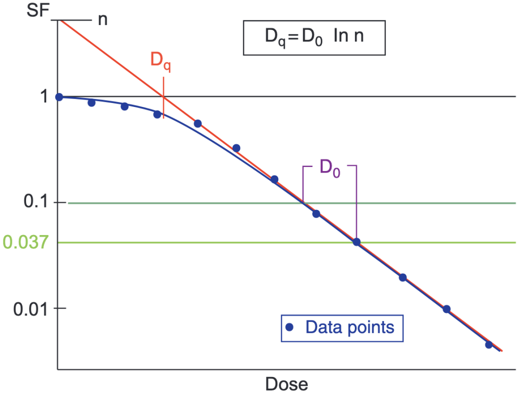


Figure: The single-hit, multitarget survival curve. Notice that it is curved at low doses and straight (e.g., survival is an exponential function of dose) at high doses. This allows you to calculate D_0 by taking measurements in the high-dose region

- The high-dose portion of the curve is actually a straight line:

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
  - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
  - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Important to note

Do not measure D_0 at SF $= 0.37$ if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between SF $= 0.1$ and SF $= 0.037$.

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
  - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Important to note

Do not measure D_0 at SF $= 0.37$ if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between SF $= 0.1$ and SF $= 0.037$.

  - D_0 is a measure of the cell's inherent radiation sensitivity. Most D_0 values are somewhere around 1 Gy.

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
  - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Important to note

Do not measure D_0 at SF = 0.37 if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between SF = 0.1 and SF = 0.037.

  - D_0 is a measure of the cell's inherent radiation sensitivity. Most D_0 values are somewhere around 1 Gy.
  - $D_{10} = 2.3 \times D_0$ = a dose that will reduce surviving fraction by tenfold (i.e., from SF = 0.1 to SF = 0.01).

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
  - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Important to note

Do not measure D_0 at SF $= 0.37$ if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between SF $= 0.1$ and SF $= 0.037$.

  - D_0 is a measure of the cell's inherent radiation sensitivity. Most D_0 values are somewhere around 1 Gy.
  - $D_{10} = 2.3 \times D_0 =$ a dose that will reduce surviving fraction by tenfold (i.e., from SF $= 0.1$ to SF $= 0.01$).
- The low-dose portion of the curve is known as the "shoulder":

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
  - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Important to note

Do not measure D_0 at SF $= 0.37$ if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between SF $= 0.1$ and SF $= 0.037$.

  - D_0 is a measure of the cell's inherent radiation sensitivity. Most D_0 values are somewhere around 1 Gy.
  - $D_{10} = 2.3 \times D_0 =$ a dose that will reduce surviving fraction by tenfold (i.e., from SF $= 0.1$ to SF $= 0.01$).
- The low-dose portion of the curve is known as the "shoulder":
  - D_q tells you how wide the shoulder is.

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
  - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Important to note

Do not measure D_0 at SF $= 0.37$ if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between SF $= 0.1$ and SF $= 0.037$.

  - D_0 is a measure of the cell's inherent radiation sensitivity. Most D_0 values are somewhere around 1 Gy.
  - $D_{10} = 2.3 \times D_0 =$ a dose that will reduce surviving fraction by tenfold (i.e., from SF $= 0.1$ to SF $= 0.01$).
- The low-dose portion of the curve is known as the "shoulder":
  - D_q tells you how wide the shoulder is.
  - D_q is a measure of the cell's repair capacity. More repair means a larger D_q and a larger shoulder

Single-Hit, Multitarget Model

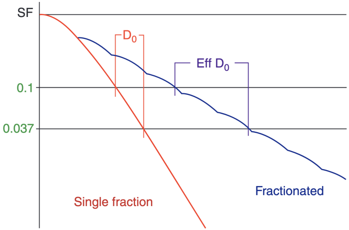
Advantages and Disadvantages

Fractionated Radiation and Effective D_0

- A fractionated radiation survival curve has a shallower slope compared to a single-fraction survival curve.

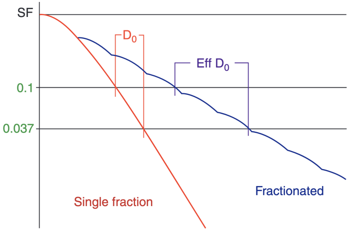
Fractionated Radiation and Effective D_0

- A fractionated radiation survival curve has a shallower slope compared to a single-fraction survival curve.
- Assuming complete repair in between fractions, the shoulder is repeated with each fraction.



Fractionated Radiation and Effective D_0

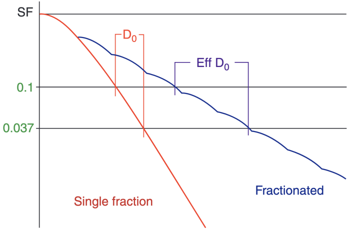
- A fractionated radiation survival curve has a shallower slope compared to a single-fraction survival curve.
- Assuming complete repair in between fractions, the shoulder is repeated with each fraction.
- D_0 versus effective D_0. When radiation is fractionated, you need much more dose to achieve the same amount of cell kill. Therefore, effective D_0 is always larger than D_0



Fractionated Radiation and Effective D_0

Continue

- The slope of a fractionated survival curve is called effective D_0.



Fractionated Radiation and Effective D_0

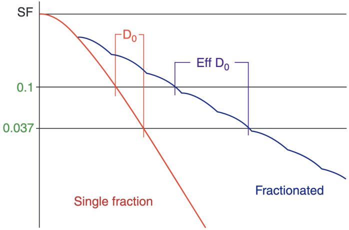
Continue

- The slope of a fractionated survival curve is called effective D_0.
- For a single fraction of D Gy with a surviving fraction of SF_D:

$$SF_D = e^{-D/D_0}$$

$$SF_D = e^{-D/\text{effective } D_0}$$

$$\text{effective } D_0 = -\frac{D}{\ln(SF_D)}$$



- The slope of a fractionated survival curve is called effective D_0.
- For a single fraction of D Gy with a surviving fraction of SF_D:

$$SF_D = e^{-D/D_0}$$
$$SF_D = e^{-D/\text{effective } D_0}$$
$$\text{effective } D_0 = -\frac{D}{\ln(SF_D)}$$

-

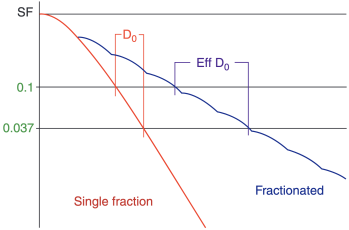


Table of Contents

Table of Contents

Table of Contents

Table of Contents